

PROBLEM STATEMENT #64 | Sustainability & Green Tech

Disaster Relief Supply & Volunteer Coordinator

1. Problem Context & Overview

An emergency disaster management platform mapping critical shelter supply needs (water, blankets, medical kits), dispatching volunteer skill sets, and tracking relief vehicle deliveries.

Target Stakeholders / Actors: *Relief Volunteer, Disaster Manager*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]

Description: The system shall match incoming field shelter supply requisitions against available central warehouse supplies and allocate delivery priority.

Sample Acceptance Criteria: Pass: Critical need given highest dispatch priority and routed to logistics manifest; Fail: Low priority item dispatched ahead of water/medical needs.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]

Description: The disaster mapping dashboard must operate with low network bandwidth consumption and support offline GIS data caching.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

- **UML Use-Case Diagram:**

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

- **Use-Case Flow Specification:**

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.